

Lab 3 Tasks.

Understanding and Practicing CMMI Level 1, 2 and 3

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Activity 1: Sample Requirements Document

1. Sample Requirements Document

Project: Library Management System

Prepared for: Lab 03 – Activity 1

Purpose: This document lists the basic requirements for a system that lets a librarian issue and track books, and lets members view their borrowing activity online.

ID	Requirement	Ambiguity / Issue Found	Priority
R1	The system shall allow a librarian to issue a book to a registered member.	Does not state what happens if the member already has an overdue book.	High
R2	The system shall calculate a late fee for overdue books.	The fee amount or calculation formula (per day/per week) is not mentioned.	High
R3	The system should respond fast when searching for a book.	"Fast" is not measurable — no target response time is given.	Medium
R4	Members shall be able to check their borrowing history online.	No mention of how a member logs in or is authenticated.	High
R5	The system shall send a notification before a due date.	Notification channel (email/SMS/app) and timing (how many days before) are not specified.	Medium
R6	The librarian can manage the system.	"Manage" is undefined — unclear exactly which actions a librarian is allowed to perform.	Low

2. Ambiguities, Missing Requirements & Inconsistencies Identified

- Ambiguous wording: R3 uses the vague word "fast" instead of a measurable response time (e.g., within 2 seconds).
- Missing detail: R2 does not define how the late fee is calculated (rate per day, maximum cap, etc.).
- Missing requirement: There is no requirement covering login/authentication, even though R4 assumes a member can securely view personal records.
- Missing detail: R5 does not specify the notification channel or how many days in advance it should be sent.
- Vague scope: R6 says the librarian can "manage the system" without listing which specific actions (add member, remove book, edit records) are allowed.

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- Missing requirement: There is no rule for what happens if a member already has an overdue book and tries to borrow another (R1).

Explanation

These issues are common in Level 1 (Initial) organizations. Because requirements are gathered informally and not reviewed against a checklist, vague terms ("fast", "manage") and missing details slip through. This is exactly why CMMI introduces the Requirements Management (REQM) process area at Level 2 — to make sure requirements are written down, reviewed, and traceable before development starts.

Activity 2: Student Grading System Scenario

Scenario: A small software company is developing a student grading system. The team works quickly to meet deadlines, but the following issues are observed: developers patch the live system directly, requirements are never documented, testing only starts after complaints, each developer uses a different Sami U

Task 1: Identify the Level 1 Processes Present in the Scenario

At least four Level 1 symptoms are present in the scenario. Each one is identified and explained below.

Observed Practice	Level 1 Symptom	Explanation
Developers update the live/production system directly whenever a bug is reported.	Chaotic Process	There is no defined release or change-control procedure, so fixes go straight into production without review, approval, or testing.
Requirements are discussed verbally in meetings but never written down.	Lack of Standardization	With nothing documented, there is no single agreed reference for what the system must do, so understanding depends on memory and differs between people.
Testing only starts after customers report incorrect results.	Reactive Management	The team reacts to a problem after it has already reached the customer, instead of planning testing as part of development.
Each developer follows their own coding style and folder layout.	Lack of Standardization	There are no shared conventions, templates, or guidelines, which makes the codebase inconsistent and hard for others to maintain.
Defects are noted informally in emails with no central record.	Chaotic Process	There is no defined defect-logging process, so issues can be lost, duplicated, or simply forgotten before they are fixed.

Explanation: All five practices point back to the three defining traits of CMMI Level 1 — work is chaotic (no defined procedure for changes or defect tracking), reactive (testing only happens once a customer

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complains), and unstandardized (no shared coding conventions). None of these failures are unique to one bad employee; they happen because the organization has no defined process to fall back on, so outcomes depend entirely on individual habits.

Task 2: Suggest a QA Improvement Toward Level 2 for Each Process

For each Level 1 issue identified in Task 1, a specific Level 2 Process Area and a concrete QA mechanism is proposed below to help the team move toward Managed (Level 2) practices.

Level 1 Issue (from Task 1)	Level 2 Process Area	Suggested QA Improvement
Direct changes made to the live system.	CM – Configuration Management	Introduce version control with a review/approval step before any change is merged or released.
Requirements only discussed verbally, never documented.	REQM – Requirements Management	Record requirements in a shared document and require sign-off from the team before development begins.
Testing only happens after customer complaints.	PMC – Project Monitoring & Control	Schedule testing as a planned activity within the project timeline, tracked and checked before release.
Inconsistent coding style and folder structure.	PPQA – Process and Product Quality Assurance	Create a basic coding-standard and folder-structure checklist that every developer follows and is periodically audited.
Defects tracked informally through email.	MA – Measurement and Analysis	Maintain a central defect log (spreadsheet or tracking tool) so defects can be counted, analyzed, and followed up.

Explanation: Each fix is deliberately small and practical — a checklist, a log, or a simple review step — rather than a large process overhaul. This matches how CMMI Level 2 works in practice: it does not require an organization-wide standard yet (that only comes at Level 3); it simply asks each project to put basic, repeatable controls in place so that outcomes become predictable instead of accidental.

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Homework:

EduTech Solutions — Multi-Level Scenario Analysis & Process Transformation

Scenario: EduTech Solutions, a vendor of library and administrative software, is experiencing project delays, frequent post-release bugs, high staff turnover, and inconsistent code quality across its teams. The five practices below were observed in the organization.

Task 1: Defect & Process Identification

Each of the five observed practices is mapped to its CMMI Level 1 symptom — Chaotic, Reactive, or Lack of Standardization — along with a short explanation.

Observed Practice	**CMMI Level 1 Symptom**	**Explanation**
A new feature requested by a client is coded immediately, without checking if it clashes with existing functionality.	Reactive / Chaotic	There is no change-review step, so the team acts on requests right away instead of assessing impact first — this risks breaking working features.
Project managers set delivery dates based on personal experience rather than past project data.	Chaotic	Without recorded metrics from earlier projects, estimates are guesses rather than data-driven, which explains the frequent delays.
Every developer names files and organizes folders their own way; work becomes hard to maintain once that person leaves.	Lack of Standardization	No common convention exists across the team, so project knowledge lives inside individual habits instead of shared documentation.
Subcontractors are brought on quickly with no formal check of their work quality.	Chaotic / Reactive	There is no supplier-evaluation step, so poor-quality deliverables are usually discovered only after integration, when it is costlier to fix.
Before release, testing is just a short manual check done by the developer themselves.	Reactive / Lack of Standardization	Testing has no plan, checklist, or independent review — it happens at the very last moment rather than throughout development.

Explanation: The pattern here is the same as in any Level 1 organization: there is no institutional memory. Decisions (estimation, hiring subcontractors, naming files) are made by individual judgment in the moment, and quality is checked, if at all, only at the very end. This is why EduTech sees repeated, similar problems on every project instead of learning from the last one.

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Task 2: Level 2 (Managed) Remediation Plan

Five project-level QA controls are proposed below, selected from REQM, PP, PMC, CM, PPQA, SAM, and MA, to help EduTech Solutions transition from Level 1 to Level 2.

#	Identified Failure	Proposed Level 2 Process Area	Concrete QA Mechanism
1	Features implemented without checking conflicts with existing code.	REQM (Requirements Management)	A Change Request Form and a Requirements Traceability Matrix, so every new request is reviewed and approved before coding starts.
2	"Gut feeling" project timeline estimates.	PP (Project Planning)	An Estimation Worksheet that records effort, resources, and past project data to support a documented, realistic schedule.
3	Inconsistent folder structures and naming conventions.	CM (Configuration Management)	A Configuration Management Plan defining a standard repository layout and naming rules, enforced through version control.
4	Subcontractors hired with no formal quality checks.	SAM (Supplier Agreement Management)	A Supplier Evaluation Checklist and a Deliverable Acceptance Log used before accepting subcontracted work.
5	Only a quick manual test walkthrough before release.	PPQA (Process and Product Quality Assurance)	A Test Case Checklist and a Release QA Sign-off Log completed before any deployment.

How Each Control Helps

1. REQM – Requirements Management

A Change Request Form forces every new feature request through a short review before coding begins, so conflicts with existing functionality are caught early instead of after deployment.

2. PP – Project Planning

Recording effort and resource data from past projects turns estimation from a guess into a documented, defensible plan, which directly reduces the delays EduTech is experiencing.

3. CM – Configuration Management

A shared repository structure and naming standard means that when a developer leaves, their code is still organized in a way the rest of the team understands — solving the turnover/knowledge-loss problem

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4. SAM – Supplier Agreement Management

Evaluating subcontractors against a checklist before accepting their work stops poor-quality deliverables from being integrated into the product in the first place.

5. PPQA – Process and Product Quality Assurance

A documented test checklist and sign-off step ensure release decisions are based on evidence that the software actually works, not just a quick manual glance.

Expected Improvement After Reaching Level 2

Before (Level 1)	**After (Level 2)**
Features coded without checking impact	Change requests reviewed and approved first
Timelines based on guesswork	Timelines based on documented estimates
Different folder/naming styles per developer	One standard project structure for everyone
Subcontractor work accepted without checks	Subcontractor work formally evaluated first
Testing squeezed in right before release	Testing planned and tracked throughout the project

Final Transformation

The overall improvement can be summarized as:

CMMI Level 1 – Initial

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Identify Process Failures

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Introduce Project-Level Controls

REQM + PP + CM + SAM + PPQA

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CMMI Level 2 – Managed

Conclusion

EduTech Solutions currently shows the classic signs of CMMI Level 1: chaotic execution, reactive problem-solving, and no standardization across teams. None of the five observed failures require a large or expensive fix — each can be addressed with a simple, concrete QA mechanism such as a checklist, a log, or a short review step. Taken together, introducing REQM, PP, CM, SAM, and PPQA moves EduTech from relying on individual heroics toward planned, repeatable, project-level control — which is the essence of CMMI Level 2 (Managed).

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